

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT II BJT AMPLIFIERS

Degree, Semester & Branch: III Semester B.E.ECE B

Course Code & Title: EC8351 Electronic Circuits I

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: Cascade and Cascode configurations

Name of the Innovative Practice: Simulation tool

Date & Duration: 17.08.2019 & 10 Minutes

ICT Tool Used: Simulation

Description:

Simulation is the imitation of the operation of a real-world process or system over time. The act of simulating something first requires that a model be developed; this model represents the key characteristics or behaviours of the selected physical or abstract system or process. The model represents the system itself, whereas the simulation represents the operation of the system over time.

Goals (Learning Outcomes):

- Students will be able to understand the concepts of cascade and cascode configurations clearly after using simulation tool.
- Students will be able to differentiate the single stage and multistage amplifiers.

Use of appropriate methods:

Justification for choosing the topic using simulation tool activity:

The simulation tool creates interest among the students rather than performing in hardware. The concepts of cascade and cascode configurations were clearly understood by the students after performing simulation.

Effective presentation

Implementation(Plan & Execution) with Proof:

The students were asked to come to Circuits and Simulation Laboratory after explaining the concepts of simulation and about the cascade and cascode configurations in the class room. The students performed simulations in the individual system and they verified the frequency response of cascade and cascode amplifiers.

Innovative Teaching Method Execution
“Cascade and Cascode configurations” using simulation tool



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students executed the results using simulation.
- ✓ During practical lab and in university lab, the students performed the simulation part correctly.
- ✓ The students understood the concepts of cascade and cascade configurations after performing simulation.

Reflective Critique:

- **Benefits:**

The students were more interested in doing the simulation activity and they clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of cascade and cascade configurations more clearly by performing simulation tool. They asked more doubts in the topic and learnt many things.

- **Challenges:**

Basically some time is needed to explain the basic concepts of simulation tool. It was somewhat challenging to explain to the slow learners initially, later on after explaining the same through system, it was understood by the students clearly.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	3	2	-	3	-	-	1	-	2	-	1

CO2: The students will be able to analyze the small signal equivalent and design BJT amplifier circuits.

References:

1. Donald. A. Neamen, Electronic Circuits Analysis and Design, 3rd Edition, Mc Graw Hill Education (India) Private Ltd., 2010.
2. Robert L. Boylestad and Louis Nasheresky, —Electronic Devices and Circuit Theory, 11th Edition, Pearson Education, 2013.
3. Adel .S. Sedra, Kenneth C. Smith, “Micro Electronic Circuits”, 6th Edition, Oxford University Press, 2010.
4. https://en.wikipedia.org/wiki/Simulation_software

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Innovative Practices Description

UNIT III SINGLE STAGE FET, MOSFET AMPLIFIERS

Degree, Semester & Branch: III Semester B.E.ECE B

Course Code & Title: EC8351 Electronic Circuits I

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: JFET & MOSFET

Name of the Innovative Practice: Mindmap

Date & Duration: 06.09.2019 & 10 Minutes

ICT Tool Used: Smart Class Room

Description:

A mind map is a diagram used to visually organize information. A mind map is hierarchical and shows relationships among pieces of the whole. It is often created around a single concept, drawn as an image in the center of a blank page, to which associated representations of ideas such as images, words and parts of words are added. Major ideas are connected directly to the central concept, and other ideas branch out from those major ideas.

Mind maps can also be drawn by hand, either as "rough notes" during a lecture, meeting or planning session, for example, or as higher quality pictures when more time is available. Mind maps are considered to be a type of spider diagram. A similar concept in the 1970s was "idea sun bursting".

Goals (Learning Outcomes):

- Students were able to understand the concepts of small signal hybrid pi model for JFET and MOSFET clearly by mind map.
- Students will be able to differentiate the BJT and FET amplifiers.

Use of appropriate methods:

Justification for choosing the topic using mind map activity:

The mind map creates interest among the students rather than asking questions. The concepts of JFET and MOSFET amplifiers were clearly understood by the students after performing mindmap activity.

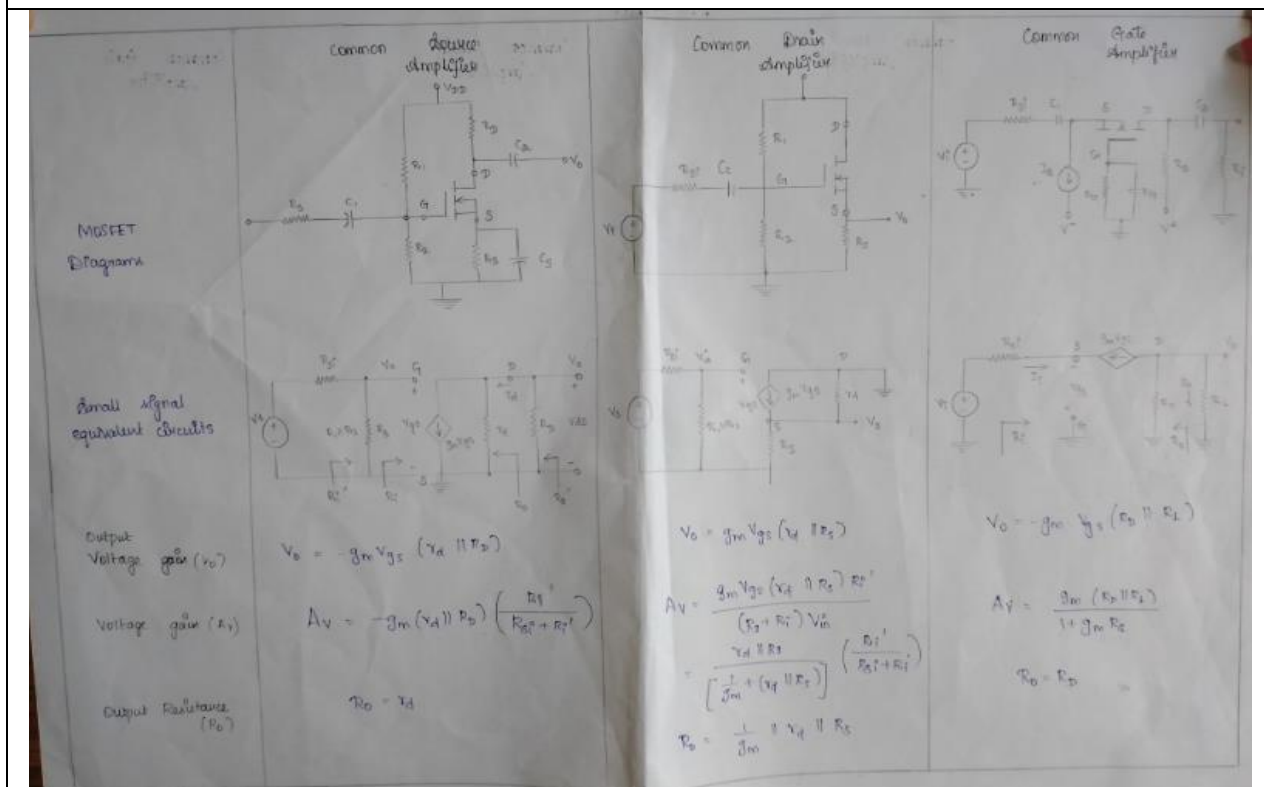
Effective presentation

Implementation(Plan & Execution) with Proof:

The students were asked to come and write on the board one by one to remember and write and draw the concepts about JFET and MOSFET amplifiers. The students performed the mindmap activity very interestingly and learnt the concepts of JFET and MOSFET.

Innovative Teaching Method Execution

“Mindmap on JFET & MOSFET”



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students understood the concepts of JFET and MOSFET.
- ✓ Most of the students performed the questions on JFET and MOSFET well that have been asked in internal assessment test.

Reflective Critique:

• **Benefits:**

The students were more interested in doing the mindmap activity and they clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of JFET and MOSFET more clearly by performing mindmap activity. The students could recollect the points and diagrams that have been taught in the class room.

• **Challenges:**

Initially some time is required by the students to remember the points that have been taught inside the class room. Some slow learners hesitated to come and write on the board. It was challenging to ask them to write on the board.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	3	2	-	3	-	-	1	-	2	-	1

CO3: The students will be able to analyze the small signal equivalent and design JFET and MOSFET amplifier circuits.

References:

1. Donald. A. Neamen, Electronic Circuits Analysis and Design, 3rd Edition, Mc Graw Hill Education (India) Private Ltd., 2010.
2. Robert L. Boylestad and Louis Nasheresky, —Electronic Devices and Circuit Theory, 11th Edition, Pearson Education, 2013.
3. Adel .S. Sedra, Kenneth C. Smith, “Micro Electronic Circuits”, 6th Edition, Oxford University Press, 2010.
4. David A. Bell, “Electronic Devices and Circuits”, Fifth Edition, Oxford University Press, 2008.
5. <https://simplemind.eu/how-to-mind-map/basics/>

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UNIT IV FREQUENCY RESPONSE OF AMPLIFIERS

Degree, Semester & Branch: III Semester B.E.ECE B

Course Code & Title: EC8351 Electronic Circuits I

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: Frequency Response of Amplifiers

Name of the Innovative Practice: Brainstorming

Date & Duration: 27.09.2019 & 15 Minutes

ICT Tool Used: Smart Class Room

Description:

It is a method for generating ideas to solve a design problem. It usually involves a group, under the direction of a facilitator. The strength of **brainstorming** is the potential participants have in drawing associations between their ideas in a free-thinking environment, thereby broadening the solution space.

Goals (Learning Outcomes):

- The students were able to understand the frequency response of amplifiers like BJT and FET and about the various cut off frequencies.
- The students were able to calculate the mid band gain of the transistors and cut off frequency.

Use of appropriate methods:

Justification for choosing the topic using brainstorming activity:

The brainstorming activity creates interest among the students rather than asking questions inside the classroom. The concepts of frequency response of amplifiers were clearly understood by the students after performing brainstorming activity.

Effective presentation

Implementation(Plan & Execution) with Proof:

The students were asked to discuss about the frequency response of amplifiers inside the class room. They were able to bring the concepts and ideas about the frequency response of amplifiers.

Innovative Teaching Method Execution
“ Frequency Response of Amplifiers ” using Brainstorming



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students have drawn the frequency response graph of the amplifiers correctly in the record note book.
- ✓ During practical lab and in university lab, the students performed the experiments correctly.
- ✓ The students understood the concepts of frequency response of the amplifiers clearly by using brainstorming activity.

Reflective Critique:

- **Benefits:**

The students were more interested in discussing about the frequency response of amplifiers and they clarified their doubts with the neighbouring students and the faculty. They learnt the basic concepts of frequency response of amplifiers more clearly by performing brainstorming activity. They asked more doubts in the topic and learnt many things.

- **Challenges:**

Basically some time is needed to explain the basic concepts of brainstorming activity. It was somewhat challenging to explain to the slow learners initially, later on after explaining the same it was understood by the students clearly.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	2	3	-	3	-	-	1	-	2	-	1

CO4: The students will be able to plot the frequency response of all amplifiers.

References:

1. Donald. A. Neamen, Electronic Circuits Analysis and Design, 3rd Edition, Mc Graw Hill Education (India) Private Ltd., 2010.
2. Robert L. Boylestad and Louis Nasheresky, —Electronic Devices and Circuit Theory, 11th Edition, Pearson Education, 2013.
3. Adel .S. Sedra, Kenneth C. Smith, “Micro Electronic Circuits”, 6th Edition, Oxford University Press, 2010.
4. David A. Bell, “Electronic Devices and Circuits”, Fifth Edition, Oxford University Press, 2008.
5. <https://en.wikipedia.org/wiki/Brainstorming>

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UNIT V BJT AMPLIFIERS

Degree, Semester & Branch: III Semester B.E.ECE B

Course Code & Title: EC8351 Electronic Circuits I

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: Rectifiers

Name of the Innovative Practice: Roleplay

Date & Duration: 09.09.2019 & 10 Minutes

ICT Tool Used: Smart Class Room

Description:

Act out or perform the part of a person or character, for example as a technique in training or psychotherapy. Participating in a role-playing game.

Goals (Learning Outcomes):

- The students were able to understand the concepts of various types of rectifiers and its usage in day today life.
- The students will be able to differentiate the types of rectifiers and its functions.

Use of appropriate methods:

Justification for choosing the topic using simulation tool activity:

The role play creates interest among the students rather than performing in hardware. The concepts of rectifiers were clearly understood by the students after role play activity.

Effective presentation

Implementation(Plan & Execution) with Proof:

Some students were asked to come in front of the class. The students were assigned the various parts involved in the rectifiers and they were asked to perform each function.

Innovative Teaching Method Execution

Roleplay on “Rectifiers”



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students were asked about the concept of rectifiers, they were able to answer well.
- ✓ The students understood the concepts of rectifiers and it reflected in the internal tests by securing good marks in that topic.

Reflective Critique:

- **Benefits:**

The students were more interested in doing the role play activity and they clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of rectifiers more clearly by performing role play activity. They asked more doubts in the topic and learnt many things.

- **Challenges:**

Basically some time is needed to explain the basic concepts of activity and the topic.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	1	3	-	1	-	-	1	-	2	-	1

CO5: The students will be able to design the regulated power supply, troubleshoot and analyze the faults in power supplies.

References:

1. Donald. A. Neamen, Electronic Circuits Analysis and Design, 3rd Edition, Mc Graw Hill Education (India) Private Ltd., 2010.
2. Robert L. Boylestad and Louis Nasheresky, —Electronic Devices and Circuit Theory, 11th Edition, Pearson Education, 2013.
3. Adel .S. Sedra, Kenneth C. Smith, “Micro Electronic Circuits”, 6th Edition, Oxford University Press, 2010.
4. David A. Bell, “Electronic Devices and Circuits”, Fifth Edition, Oxford University Press, 2008.
5. <https://en.wikipedia.org/wiki/Role-playing>